

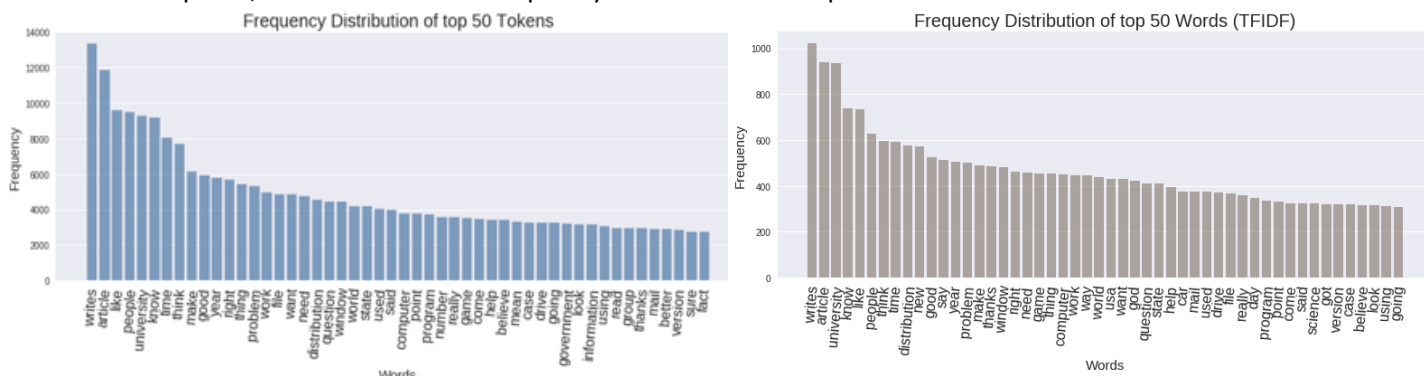
20 Newsgroup Dataset and Visualization

Introduction

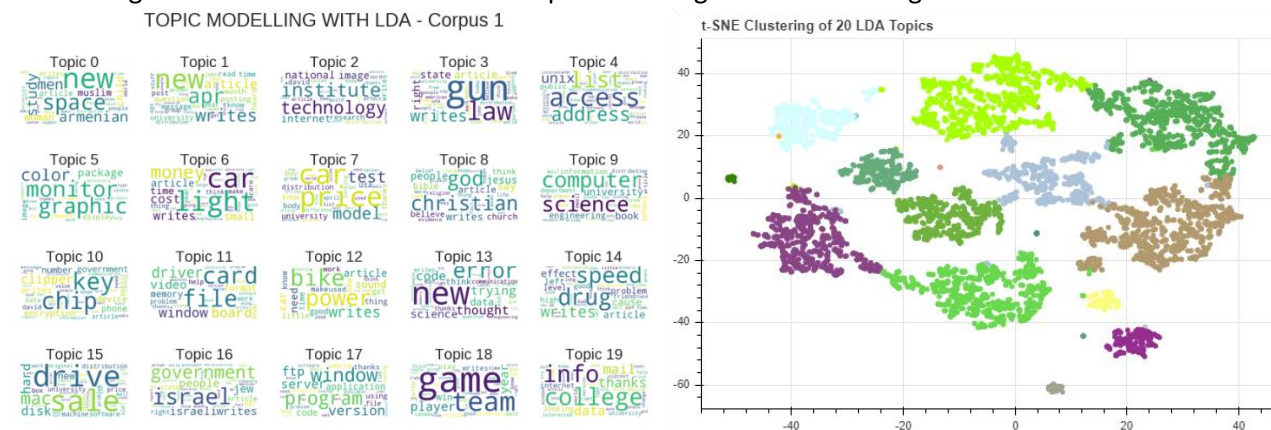
This paper explores different data visualization techniques of the popular 20 Newsgroup dataset. This corpus contained 18846 documents labeled by 20 classes which are subsets of 6 themes, with 7531964 total words and 352484 unique words across all documents. Two corpuses were built using varied preprocessing and then manipulated with natural language processing, clustering, and visualization. After cleaning the data set only 2156999 total words remain in the dataset. Preprocessing included work tokenizing, consolidating root words, removing all numbers, special characters, and things like stopwords.

Results and Analysis

Using 2 different preprocessing methods, the two corpuses are built. Metadata gets removed, sentences are tokenized, all words get lower case, numbers get removed, and words get consolidated into root words. Also, words with less than two character get omitted. Lastly block and stop words are removed such as *the*, *is*, *at*. The main difference between the two is that corpus 1 is built with a unigram method and corpus 2 uses bigram. Vocabularies were built with BoW and TF-IDF on the corpuses, below is the word frequency distribution of corpus 1.



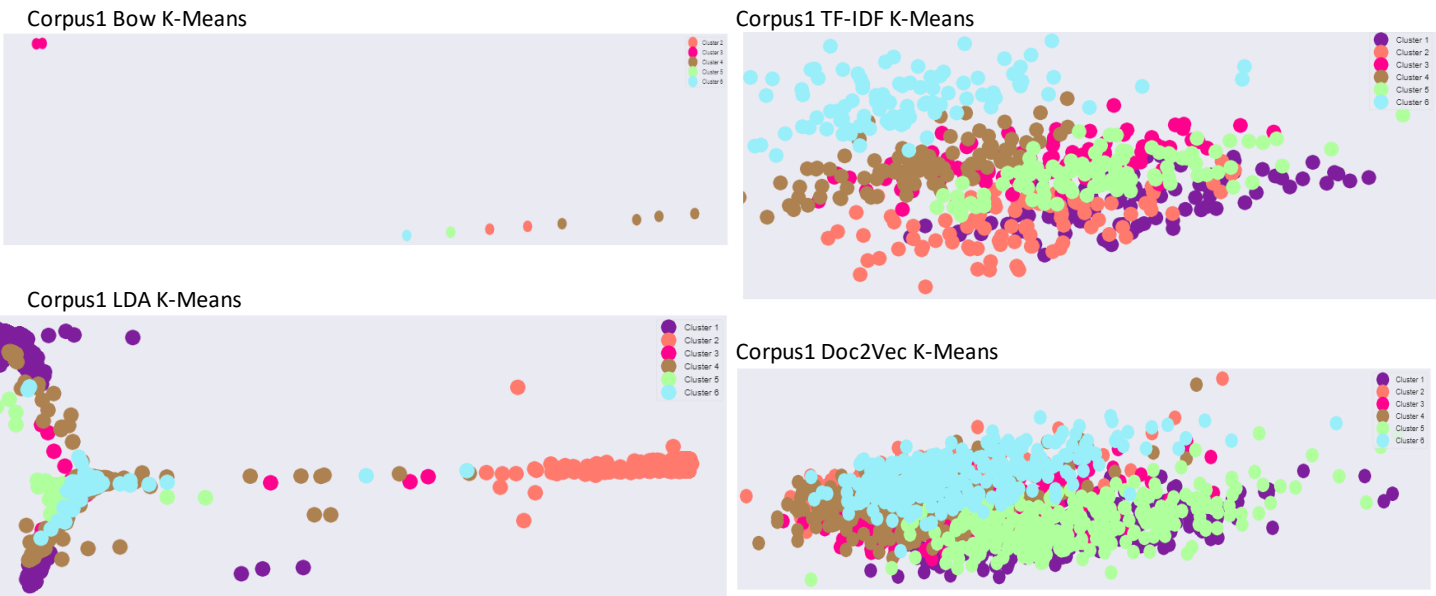
LDA from genism and sklearn was used for topic modelling and the following two visualizations were made.



Next Doc2Vec was used on these vocabularies. We used 5 different training methods and found their accuracy using logistic Regression. DBOW had the highest accuracy of the various training methods used; DBOW, DMC, DMM, DBOW+ DMC, DBOW + DMM.

Next document clustering with KMeans was performed with four different techniques; BoW, TF-IDF, LDA Topics Distributions, and Doc2Vec. The results were compared with Normalized Mutual Information (NMI) scores.

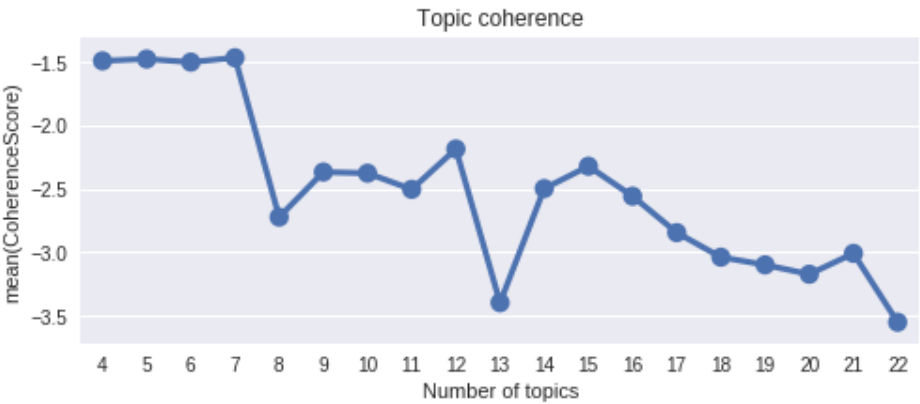
Corpus 1 Document Clustering Visualizations



NMI Scores from K-Means Document Clustering

	Corpus 1	Corpus 2
BOW	0.000421867	0.0135086
TF-IDF	0.265359	0.260429
LDA	0.257389	0.153285
Doc2Vec	0.000254634	0.000254284

Experiments were made to analyze the impact of different n-gram models in the vocabularies, different topic numbers, and different training methods for Doc2Vec. We suspect the key factor for document visualization is having sufficient features when clustering. We found the topic coherence scores for the LDA models and found this plot which shows around 13, and then 22 topics is the best number of topics for this dataset.



We were able to visualize the 20 Newsgroups datasets in various ways. Corpus 1 and 2 were made with different vocabulary building techniques. The large dataset was shown in word cloud visualizations, t-SNE mapping, intertopic distance mapping + word distribution, and sentence coloring by topic. We investigated impact of using different n-gram models, number of different topics, and different training methods for Doc2Vec. We employed the use of data manipulation tools such as Bag of Words, TFIDF, LDA for topic modeling, KMeans, Doc2Vec for analyzing the data and getting various feature representations.